EECE-477 (Power System Protection)
Fall-2025 # **Assignment**

Assignment Instructions

1. **Marks & Submission:**
   - Total marks: **35** (including **15 for class participation**).
   - **Submission is mandatory**; no other CT will be given.
   - Submit **hard copies on 19 Mar 2026**. Late submissions will not be accepted.
2. **Individual Work:**
   - This is an **individual assignment**. No discussion or communication with classmates or others is allowed.
   - For queries, meet the faculty **personally**; no class discussion will be held regarding assignment issues.
3. **Assignment Question:**
   - In the assignment, "**X**" sign should be replaced with the **last digit of your class roll number**.
4. **Report Format:**
   - Submit a **printed study report** with the solution.
   - Use **Times New Roman, font size 12**.
   - References may be included in **IEEE format**.
   - Include diagrams, tables, or figures wherever necessary.
5. **Assessment Guidelines:**

   Do's:
   - Present accurate and well-thought-out solutions.
   - Apply critical thinking in real-world scenarios.
   - Write clearly, concisely, and logically.
   - Ensure neatness and proper formatting throughout the report.
   - Prepare a professional presentation of your work.
   - Reference all sources properly using IEEE format.

   Don'ts:
   - Do not plagiarize or copy work from others.
   - Avoid unclear, incomplete, or unorganized responses.
   - Do not submit late or unformatted reports.
   - **Do not use AI-generated text without proper acknowledgment.**
   - Avoid irrelevant information or off-topic discussion.
6. **Assessment Focus:**
   - Solution accuracy and problem-solving capability.
   - Critical thinking in practical, real-world scenarios.
   - Writing quality, clarity, and organization.
   - Presentation and report formatting.
   - **Academic honesty and originality.**

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Bangladesh Rural Electrification Board (BREB) radial distribution feeder of 2 km length supplies multiple downstream loads through a step-down transformer, as shown in Fig. 1. The feeder is energized from a single source, and protection devices are required at Bus-2 and Bus-3 to ensure safe and reliable operation during fault conditions.

You have been appointed as a power system protection engineer to analyze the system, calculate fault currents, and design appropriate protection schemes including circuit breakers, current transformers (CTs), and overcurrent relays. The system parameters, transformer connection, feeder impedances, and load locations are provided in the Fig. 1. For fault analysis, load currents may be neglected.

The system base power is defined as **5X** MVA, where **X** represents the last digit of your class roll number, and appropriate base voltage levels are selected for each bus according to the system configuration.

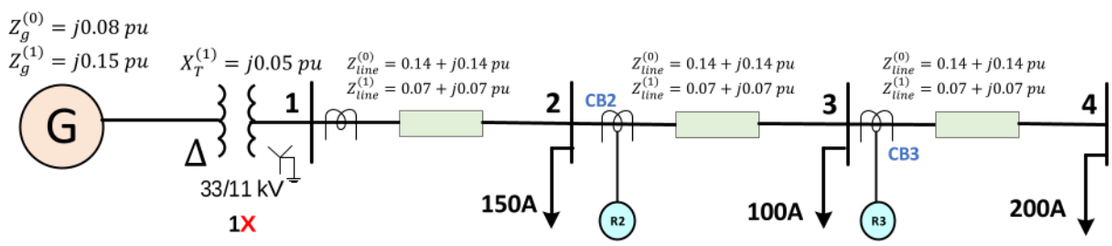

Fig. 1

Task (a): Analytical Fault Analysis
Using the given system data and selected base values, you are required to calculate the three-phase and single-line-to-ground fault currents at Bus-2 and Bus-3 in per-unit, neglecting load currents. The analysis shall be carried out by constructing the appropriate sequence networks, determining the Thevenin equivalent seen from the fault location, and computing the corresponding fault currents.

Task (b): Simulation-Based Verification
The given radial distribution system shall be modeled in a suitable power system simulation platform, and three-phase as well as single-line-to-ground faults shall be simulated at Bus-2 and Bus-3. The analytically calculated fault currents shall be verified by comparing them with simulation results, supported by appropriate plots and waveform representations.

Task (c): Actual Fault Current Calculation and Protection Equipment Selection
The per-unit fault current results obtained from analytical calculations and simulation studies shall be converted into actual fault current values using the selected base MVA. Based on the calculated line load currents at Bus-2 and Bus-3, suitable CT ratios shall be selected from the available options, ensuring adequate measurement accuracy and sufficient margin to prevent CT saturation during fault conditions. Furthermore, appropriate circuit breakers for Bus-2 (CB2) and Bus-3(CB3) shall be selected from existing options available in the Bangladesh market by considering rated voltage, continuous current capacity, and short-circuit breaking capability, with proper justification based on protection coordination, safety margins, and system reliability.

Task (d): Overcurrent Relay Design and Performance Verification
Design an overcurrent relay protection scheme for Bus-2 (R2) and Bus-3 (R3) using an IDMT or other appropriate relay characteristic. Determine the relay pickup current, time dial setting, and operating time based on the calculated fault current levels. Implement the designed relay scheme in a suitable simulation platform and verify its performance by generating time–current characteristic curves and fault response plots to demonstrate proper relay operation and coordination.